

This unit is assessed by means of a written examination paper of 1 hour 30 minutes duration. The paper will consist of objective, short-answer and long-answer questions. Students may be required to apply their knowledge and understanding of physics to situations that they have not encountered before. The total number of marks available for this examination paper is 80. It contributes 40% to IAS and 20% to the IAL in Physics.

It is recommended that students have access to a scientific calculator for this paper.

Students will be provided with the formulae sheet shown in *Appendix 5*. Any other physics formulae that are required will be stated in the question paper.

The quality of written communication will be assessed in the context of this unit through questions which are labelled with an asterisk (*). Students should take particular care with spelling, punctuation and grammar, as well as the clarity of expression, on these questions.

2.3 Waves

This topic covers the properties of different types of waves, including standing (stationary) waves. Refraction, polarisation and diffraction is also included.

This topic should be studied by exploring the applications of waves, for example, applications in medical physics or applications in music. This topic includes many opportunities to develop experimental skills and techniques.

Students will be assessed on their ability to:		Suggested experiments
28	understand and use the terms <i>amplitude</i> , <i>frequency</i> , <i>period</i> , <i>speed</i> and <i>wavelength</i>	Wave machine or computer simulation of wave properties
29	identify the different regions of the electromagnetic spectrum and describe some of their applications	
30	use the wave equation $v = f\lambda$	
31	recall that a sound wave is a longitudinal wave which can be described in terms of the displacement of molecules	Demonstration using a loudspeaker Demonstration using waves on a long spring
32	use graphs to represent transverse and longitudinal waves, including standing (stationary) waves	
33	explain and use the concepts of wavefront, coherence, path difference, superposition and phase	Demonstration with ripple tank
34	recognise and use the relationship between phase difference and path difference	
35	explain what is meant by a <i>standing (stationary) wave</i> , investigate how such a wave is formed, and identify nodes and antinodes	Melde's experiment, sonometer
36	recognise and use the expression for refractive index ${}_1\mu_2 = \sin i / \sin r = v_1 / v_2$, determine refractive index for a material in the laboratory, and predict whether total internal reflection will occur at an interface using critical angle	
37	investigate and explain how to measure refractive index	Measure the refractive index of solids and liquids